

CHAPTER - 3

INPUT DEVICES

INTRODUCTION

What is Input Device?

- Input devices are important parts of a computer that help us communicate with the system.
- These devices let us send data or commands to the computer, allowing it to process information and perform tasks.
- Whether it's typing on a keyboard or clicking a mouse, these devices enable us to interact with the computer and accomplish tasks.



TYPES OF INPUT DEVICES

- **Keyboard**

- **Pointing Device**

- a. **Mouse**

- b. **Trackball**

- c. **Joystick**

- d. **Light Pen**

- e. **Touch Screen**

- f. **Trackpad**

- **Speech Recognition**

- **Digital Camera**

- **Webcam**

- **Scanners**

- a. **Handheld scanner**

- b. **Flatbed scanner**



TYPES OF INPUT DEVICES

- **Keyboard:**
- A keyboard is the most common **data entry device**.
- Using a keyboard, the user can **type text and commands**.
- Data is entered into the computer by simply **pressing keys**.
- The number of keys on a typical keyboard varies from 84 to 104.
- The layout of a keyboard comes in various styles, such as **QWERTY, AZERTY and DVORAK**.
- **QWERTY** is the most common layout in English language computer keyboards.





POINTING DEVICE

- A **pointing device** is used to communicate with computer by pointing to locations on the monitor screen.
- Such devices do not require keying of characters, instead you can move cursor on the screen and perform move ,click or drag operations.
- Some of the commonly used pointing devices are **Mouse, Trackball, Joystick, Light Pen, Touch Screen and Trackpad.**



MOUSE

- A Mouse is a small handheld pointing device with a rubber ball embedded at its lower side and buttons on the top.
- Usually, a mouse contains two or three buttons, which can be used to input commands or information.
- It may be classified as a **mechanical mouse or an optical mouse**, based on the technology it uses.
- A **mechanical mouse** uses a rubber ball at the bottom surface, which rotates as the mouse is moved along a flat surface, to move the cursor. It is the most common and least expensive pointing device.
- An **optical mouse** uses a light beam instead of a rotating ball to detect movement across a specially patterned mouse pad.
- As the user rolls the mouse on a flat surface, the cursor on the screen also moves in the direction of the mouse's movement.



Mechanical



Optical



TRACKBALL

- A trackball is an input device is as if a mouse is turned upside down. It has a ball which can be rotated by fingers in any direction, the cursor moves accordingly.
- Unlike a traditional mouse, which you move across a surface, a trackball stays in one place, and you spin the ball to navigate on the screen.
- The size of the ball in the trackball varies from as large as a cue ball, to as small as a marble.
- Since it is a static device, instead of rolling the mouse on the top of the table the ball on the top is moved by using fingers, thumbs and palms.
- This pointing device comes with various shapes and forms but with same functions. The three shapes, which are commonly used are a **ball, button and square**.
- A lot of computer game enthusiasts and graphic designers also tend to choose to have more buttons to cut down on keyboard use.





JOYSTICK

- A joystick is a device that moves in all directions and controls the movement of the cursor.
- Joystick consists of a stick that is attached to a plastic base with a flexible rubber sheath. This plastic base houses a circuit board.
- A joystick also consists of buttons which can perform certain actions once a position on the screen has been selected using stick.
- It offers three types of control: **digital, glide and direct**. Digital control allows movement in a limited number of directions such as up, down, left and right. Glide and direct control allow movements in all directions (360°).
- It is used for **computer games**. The other applications in which it is used are flight **simulators**, training **simulators**, **CAD/CAM systems** and for **controlling industrial robots**.





LIGHT PEN

- A light pen is a pointing device that has the appearance of a pen.
- It can be used to draw on the monitor screen or to pick a menu item.
- It is a pen like device, which are connected to the machine with cable.
- In a small tube, a photocell and an optical system(lens) are housed.
- The photocell sensor element determines the screen location and sends a signal to the CPU when the tip of a light pen is moved across a monitor screen while the pen button is pressed.
- Light pens are perfect for applications where desk space is limited.

Uses of Light Pen:

- Graphic Designs,
- Highlighting,
- Painting - draw digital art with the help of a light pen



LIGHT PEN



TOUCH SCREEN:

- A touchscreen is a type of input device that allows users to interact with a digital display by directly touching the screen's surface.
- It enables the user to perform various actions, such as selecting options, typing on a virtual keyboard, drawing, or manipulating objects, by physically touching the screen.
- Typically, it is used in information-providing systems like hospitals, airlines and railway reservation counters, amusement parks, ATM and so on.



TRACKPAD

- Trackpad (also referred to as touchpad) is a pointing device that works by sensing the movement of fingers across a small sensitive surface and translating them into the pointer movement on the computer screen.
- It is generally used in laptops but can also be connected to a PC through a cord.
- It is also equipped in personal digital assistants (PDAs) and media players such as the iPod.
- Typically, a trackpad also consists of two or three buttons which work as mouse buttons.
- Many trackpads are also strike sensitive, that is, the user can tap on the trackpad to perform operations like selecting an object, maximizing/ minimizing the window, etc.



TRACKPAD



SPEECH RECOGNITION

- The user can simply instruct the computer, with the help of a microphone (along with a speech recognition software), to perform a task.
- It is the technology by which sounds, words or phrases spoken by humans are converted into digital signals, and these signals are transformed into computer-generated text or commands.
- The computer-based speech-recognition systems can be used to create text documents such as letters or e-mails, to browse the Internet, and to navigate among applications by voice commands.
- However, as compared to other input devices, the reliability of the speech recognizer is lesser. Eg. (see and sea).
- This technology is the core of voice assistants like Siri and Alexa,



DIGITAL CAMERA

- A Digital camera, stores images digitally rather than recording them on a film.
- Once a picture has been taken, it can be transferred to a computer system and then manipulated with an image editing software , and printed.
- The big advantage of digital cameras is that making photos is both inexpensive and fast because there is no film processing.



WEBCAM

- A webcam (short form of web camera) is a portable video camera, which captures live video or images that may be viewed in real time over a network or the Internet.
- It is just a small digital camera that is either built in your computer (in most laptops) or can be connected through a USB port .
- It is normally placed on top of the PC monitor or laptop to capture images of the user while he/she is working on the computer.



SCANNERS

- A scanner is an input device that converts a document (picture or text) into an electronic format that can be stored on the disk.
- The electronic image can be edited, manipulated, combined, and printed by using the image editing software.
- Scanners are also called optical scanners as they use a light beam to scan the input data.
- Moreover, they can store images in both greyscale and color mode.
- The two most common types of scanners are **Hand-Held Scanners and Flatbed Scanners**.



HAND-HELD SCANNERS

- Hand-held scanner consists of LED's, which are placed over the document to be scanned.
- This scanner performs the scanning of document very slowly from the top to the bottom with its light on. In this process, all the documents are converted and then stored as images.

FLATBED SCANNERS

- Flatbed scanner looks similar to a photocopier machine. It consists of a box containing a glass plate on its top and a lid that covers the glass plate.
- This Flatbed glass plate is used for placing the document to be scanned. The light beam is placed below the glass plate and when it is activated it moves horizontally from left to right.
- After scanning one line, the light beams moves in order to scan the next line and procedure is until all the lines are scanned.



HAND-HELD SCANNERS



FLATBED SCANNERS



CHAPTER - 4

OUTPUT DEVICE

- **Output devices** are hardware that display or produce the results of a computer's processing. They convert digital data into formats we can see, hear, or touch.
- There are two basic categories of output:
- **Hard Copy:** The physical form of output is known as hard copy. Hard copies example includes books, official letters, notes, news papers, magazines etc.
- **Soft Copy:** The electronic version of an output, which usually resides in computer memory and/or on disk, is known as soft copy. oft copies example includes ebooks, pdf files, word documents, presentation files, scanned copy etc.



CHAPTER - 4

OUTPUT DEVICE

- Based on the hard copy and soft copy outputs, the output devices are classified into hard copy and soft copy output devices.
- **Printers**
- **Plotters**, are the most commonly used hard copy output devices
- **Monitors**, are used soft copy output devices



PRINTERS

- A printer prints information and data from the computer onto paper.
- The quality of a printer is determined by the clarity of a print it can produce, that is, its resolution.
- For printers, the resolution is measured in DPI (dots per inch). The more the DPI, the better will be the quality of image.
- Printers are divided into two basic categories: Impact printers and Non-Impact printers.
- Impact printers work by physically striking a head or needle against an ink ribbon to make a mark on the paper. This includes
 - Dot Matrix printers, and Drum Printers.
 - Ink-Jet and Laser Printers are non-impact printers.
 - Hydra Printer





Increasing Dots Per Inch



DOT MATRIX PRINTERS

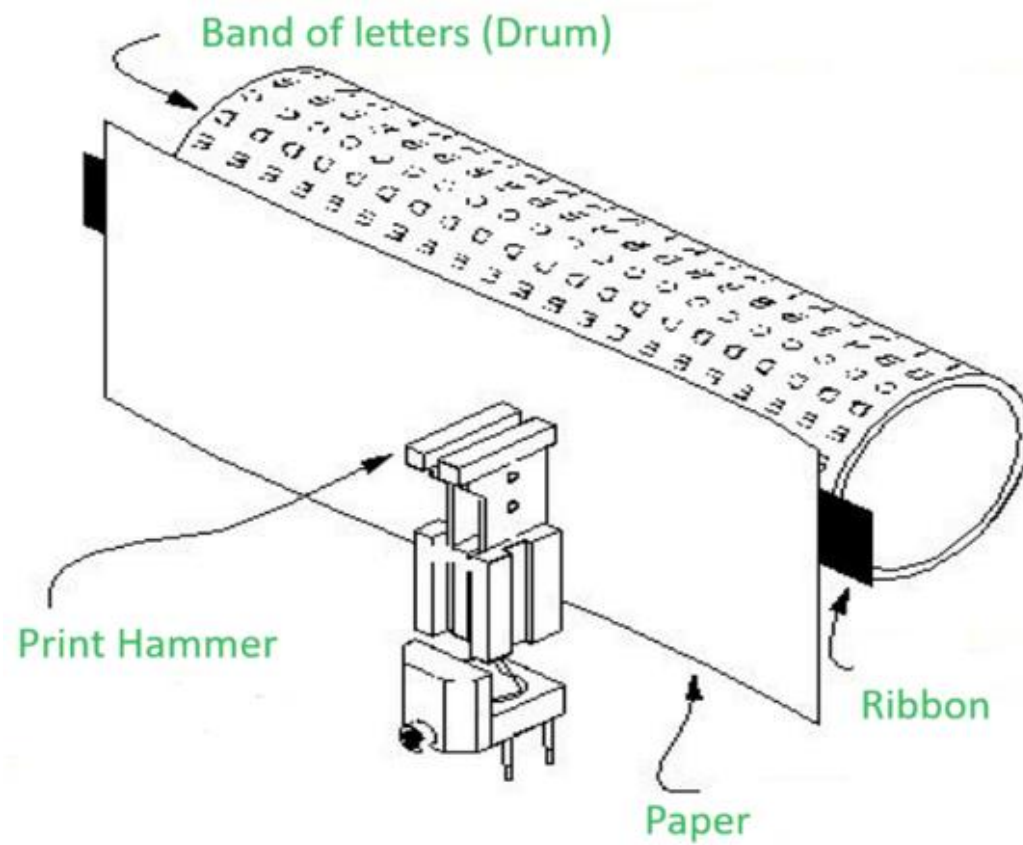
- Dot matrix printer (also known as the wire matrix printer) uses the oldest printing technology and it prints one character at a time.
- It prints characters and images as pattern of dots. The speed of dot matrix printers is measured in characters per second (cps). The speed can vary from about 200 to over 500 cps.
- The print quality is determined by the number of pins (the mechanisms that print the dots), which can vary from 9 to 24.
- Dot matrix printer are inexpensive and have low operating cost.
- Major limitation of Dot matrix printer it prints only in black and white.



DRUM PRINTER

- Drum printer is a line printer, that is, it can print a line in a single operation.
- In this, the paper is placed between the drum and the printing hammer, and the character is printed onto the paper by activating the particular hammer.
- Drum printer only prints the pre-defined characters it is restricted to printing color pictures and special types of font text.
- Its printing speed varies from 300 lines to 2000 lines per minute with 96 to 160 characters on a 15-inch line.
- Also creates noise in the environment because it uses the printing hammer to print the characters.



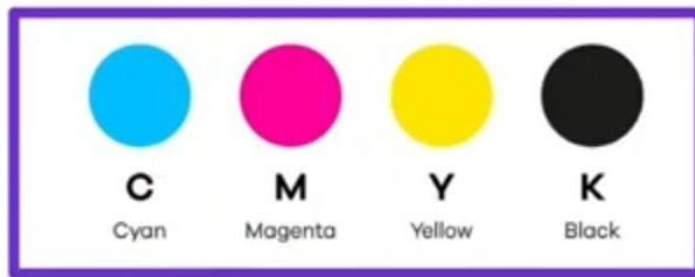


INK-JET PRINTER

- The most common type of printer found in homes today is the ink-jet printer.
- An ink-jet printer is a printer that places or spray extremely small droplets of ink onto paper to create an image. Being a non-impact printer, it does not touch the paper while creating an image.
- Instead, it uses a series of nozzles to spray drops of ink directly onto the paper. Inkjets were originally manufactured to print in monochrome (black and white) only.
- However, the print head has now been expanded and the nozzles increased to accommodate cyan (C), magenta (M), yellow (Y) and black (K).
- This combination of color's is called CMYK.



- Ink-jet printers are costlier than dot matrix printers and the quality is much better.
- These printers can print any shape of character, which a user can specify. This allows the printer to print many special characters, different sizes of prints, and enables it to print graphics such as charts and graphs.
- Ink-jet printer typically print with a resolution of 600 dpi or more.



LASER PRINTER:

- It is a very fast printer, which operates on the same principle as that of a photocopy machine.
- Most laser printers can print text and graphics with a very high-quality resolution.
- They are also known as page printers because they process and store the entire page before they actually print it.
- They produce sharp, crisp images of both text and graphics, providing resolutions from 300 to 2400 dpi. Today, the resolution of most printers is 600 dpi.
- They are quiet and fast, can print 4–32 text-only pages per minute for individual microcomputers and up to 200 pages per minute for mainframes.
- Laser printer can print in excess of 2000 lines per minute.
- Often faster than ink-jet printers but are more expensive to buy and maintain than the other printers



HYDRA PRINTER

- Hybrid document reproduction apparatus (HYDRA) printer, popularly known as one printer, is a device that consolidates the capabilities of multiple devices in one machine
- It may include some or all of the devices like printer, scanner, photocopier and fax machine.
- Apart from these devices, some hydra printers contain memory card slots which facilitate easier printing of photos and also have the in-built wireless capabilities that make haring of this printer with other systems easier.



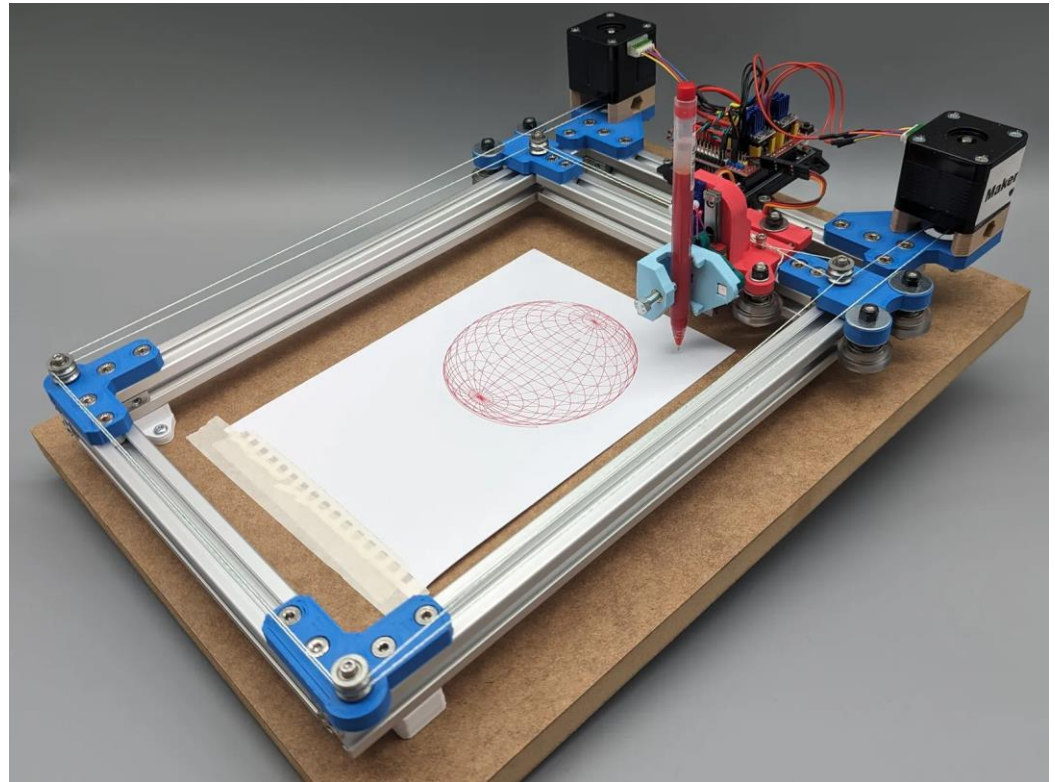
PLOTTER

- A plotter is a pen-based output device that is attached to a computer for making vector graphics , that is, images created by a series of many straight lines.
- It is used to draw high-resolution charts, graphs, blueprints, maps, circuit diagrams and other line-based diagrams.
- It is similar to a printer, but it draws lines using a pen.
- Plotters are relatively expensive as compared to printers but can produce more printouts than standard printers.
- Multicolor plotter uses different colored pens to draw different color. Color plots can be made by using four pens (CMYB) no human intervention to change them.
- They are mainly used for Computer Aided Design (CAD) and Computer Aided Manufacturing (CAM) applications such as printing out plans for houses or car parts.



There are two different types of plotters:

- **Drum plotter** (where the paper moves) and
- **Flatbed plotter** (where the paper is stationary)



DRUM PLOTTER

- Drum plotters, as name suggests, they use a rotating drum to move the paper left and right while one or more fixed pens write up and down.
- Drum plotters are used to produce continuous output such as plotting earthquake activity or for long graphic output such as tall building structures.



FLATBED PLOTTER

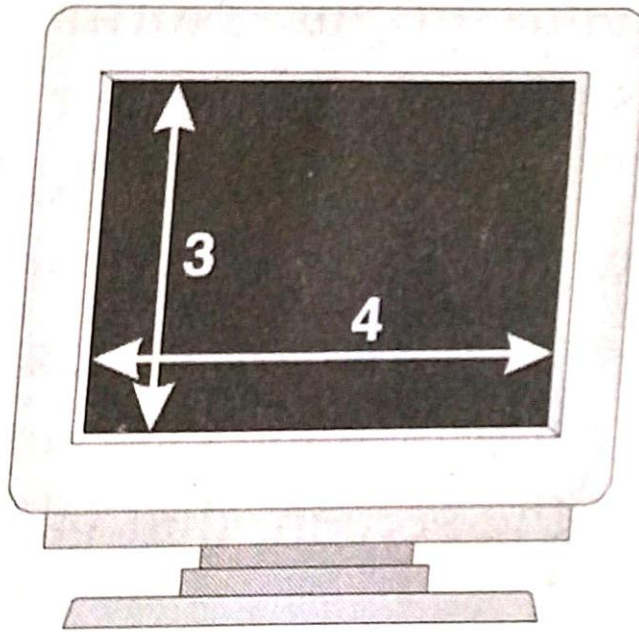
- Flatbed plotters consist of a stationary, horizontal plotting surface on which paper is fixed.
- The pen is mounted on a carriage, which can move horizontally, vertically, leftwards or rightwards to draw lines. In flatbed plotters, the paper does not move, the pen-holding mechanism provides all the motion.



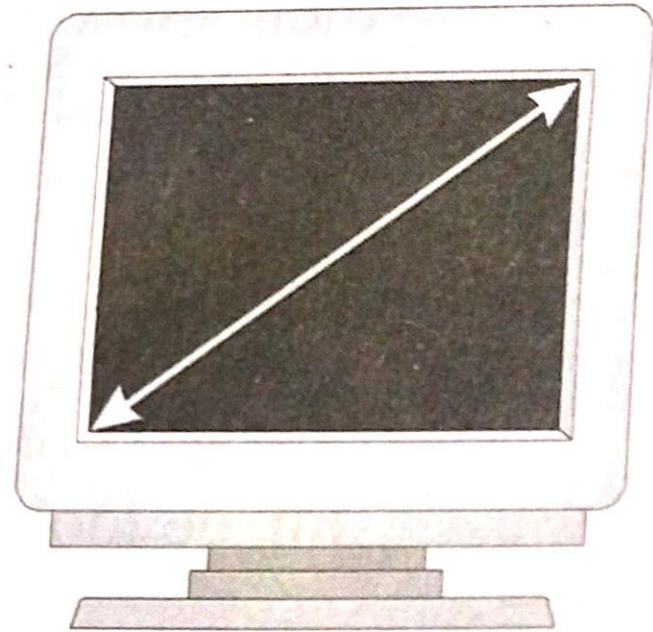
MONITOR

- A computer monitor is a TV-like display attached to the computer on which the output can be displayed and viewed.
- The computer monitor can either be a monochrome display or a color display.
- A monochrome screen uses only one colour (usually white, green, amber or black) to display text on contrasting background.
- The size of the display is described based on two parameters:
Aspect ratio and Screen size.
- Aspect ratio is the ratio of the width of the display screen to the height, that is, the ratio of vertical points to the horizontal points necessary to produce equal-length lines in both directions on the screen.





(a) Aspect Ratio



(b) Screen Size





Monochrome display



MONITOR

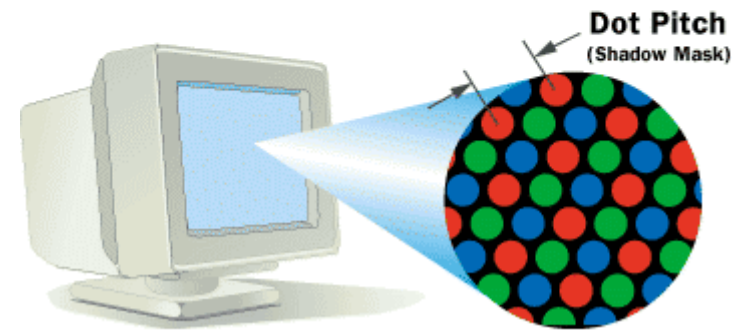
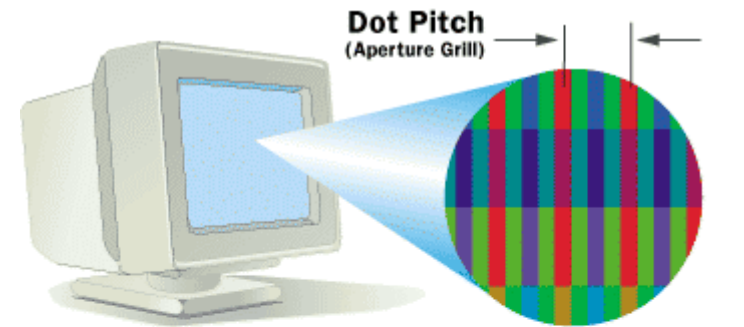
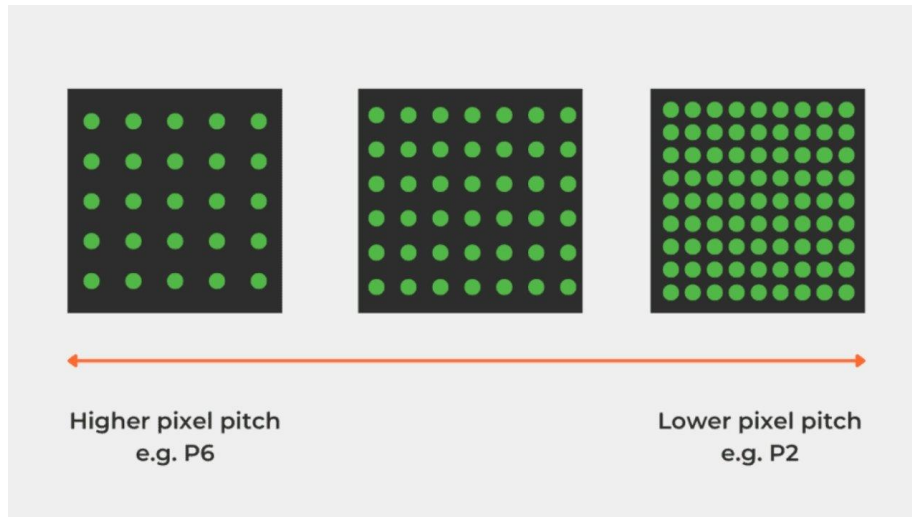
- Sometimes, while watching television, you may notice that the picture looks a bit blurred. The reason behind this is that the displayed image is not solid but is created by the configurations of dots. These dots are known as picture elements, pels, or simply pixels. The golden rule of a sharp image is that the more the pixels, the sharper the picture.
- The screen clarity depends on three basic qualities:
- **Resolutions, Dot Pitch & Refresh Rate**
- **Resolutions:** It refers to the number of pixels in the horizontal and vertical directions on the screen.
- **Dot Pitch:** It is the measurement of the diagonal distance between two like-coloured (red, green or blue) pixels on a display screen. It is measured in millimeters.
- A smaller dot pitch means pixels are closer together, resulting in a sharper image, while a larger dot pitch can lead to a less defined picture.





Resolutions





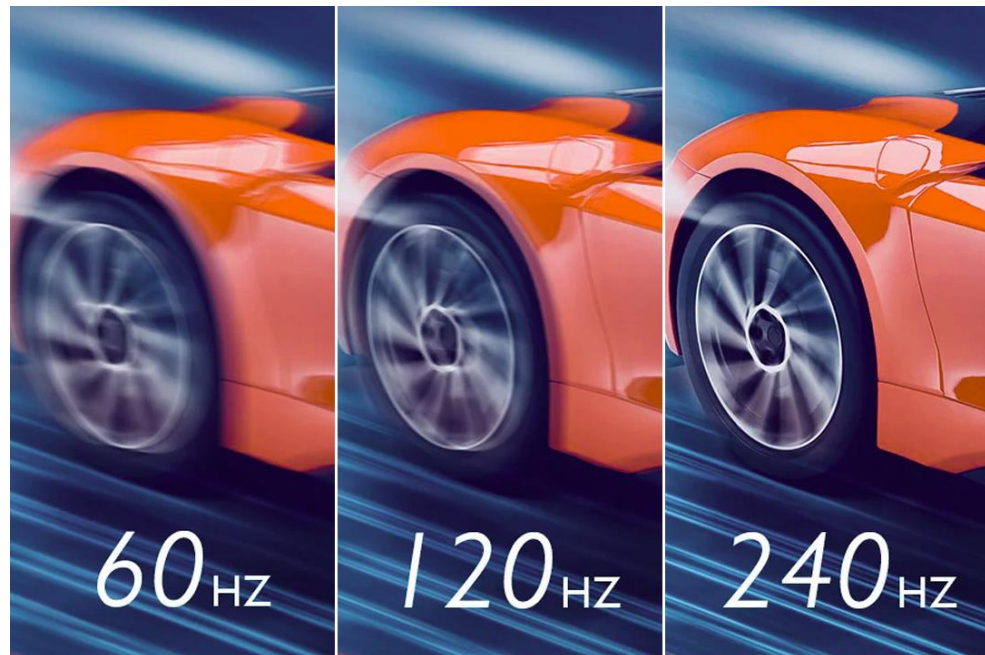
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Dot Pitch



MONITOR

- **Refresh Rate:** It is the number of times per second the pixels are recharged so that their glow remains bright.
- The refresh rate for a monitor is measured in Hertz (Hz) and varies from 60 to 75 Hz. A refresh rate of 60 Hz means image is redrawn or display updates the image on the screen 60 times a second.
- The higher the refresh rate, the more solid the image looks on the screen, it does not flicker.



MONITOR

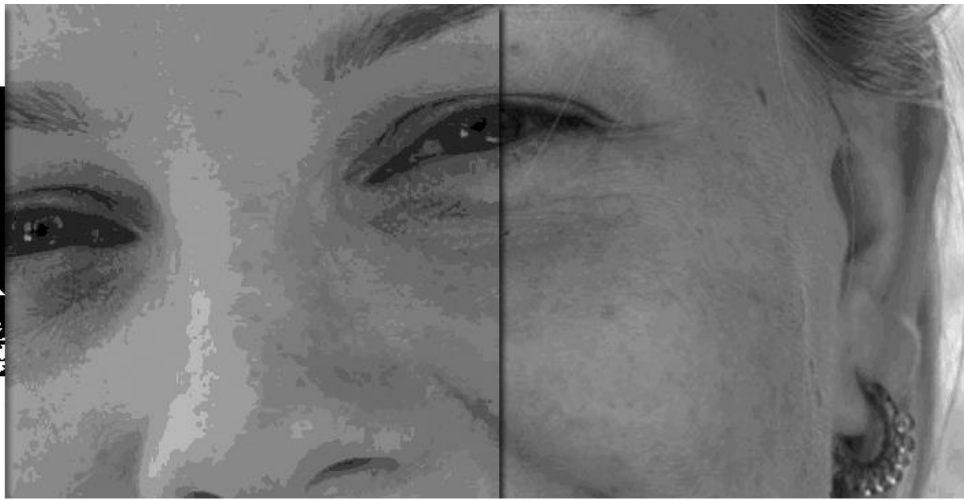
- **Colour Depth:** Colour depth, also referred to as **bit depth**, refers to the number of bits assigned to each pixel in the image and the number of colours that can be created from those bits. In simple words, it refers to the number of colours that a monitor can display.

2^1		→ 1 bit	→ 2 colors
2^2		→ 2 bit	→ 4 colors
2^3		→ 3 bit	→ 8 colors
2^4		→ 4 bit	→ 16 colors
2^5		→ 5 bit	→ 32 colors
2^6		→ 6 bit	→ 64 colors
2^7		→ 7 bit	→ 128 colors
2^8		→ 8 bit	→ 256 colors
2^{16}		→ 16 bit	→ 32,768 colors
2^{24}		→ 24 bit	→ 16,777,216 colors





1-bit



4-bit

8-bit



TWO TYPES OF MONITOR

- Cathode Ray Tube Monitor
- Liquid crystal display monitors (LCD)



CATHODE RAY TUBE MONITOR



LIQUID CRYSTAL DISPLAY MONITORS (LCD)



CATHODE RAY TUBE MONITOR

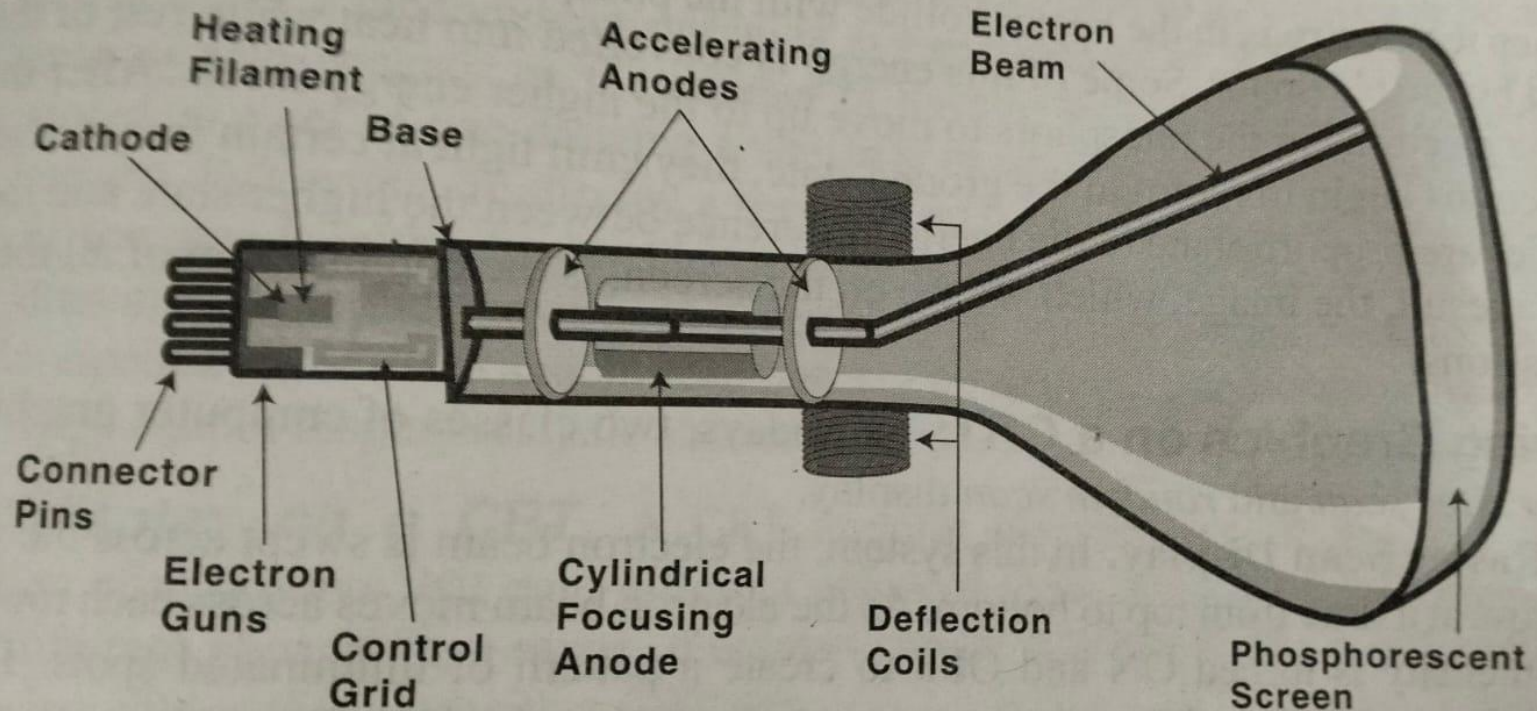


Figure 4.45 *Cathode Ray Tube*



CATHODE RAY TUBE MONITOR

- The basic operations of these Cathode Ray Tube is similar to that in television sets.
- A beam of electrons (cathode rays) emitted by an electron gun passes through focusing and deflection systems that direct the beam toward specified positions on the phosphor-coated screen.
- The phosphor then emits a small spot of light at each position contacted by the beam.
- The primary components of an electron gun in a CRT are the heated metal cathode and a control grid.
- Heat is supplied to the cathode by directing a current through a coil of wire, called the filament, inside the cylindrical cathode structure.



CATHODE RAY TUBE MONITOR

- In the vacuum inside the CRT envelope, the free, negatively charged electrons are then accelerated toward the phosphor screen.
- Before reaching the phosphor-coated screen, the electrons have to be passed through monitor's focusing system. It focus the electron flow into a very thin beam and then in a specific direction



CATHODE RAY TUBE MONITOR

DISPLAYING GRAPHICS ON A CRT:

Two classes of computer graphic display *Raster Scan Display* and *Random Scan Display*.

COLOUR DISPLAY ON CRT

The two basic techniques for producing colour displays with a CRT are:
Beam Penetration and *Shadow Masking*



DISPLAYING GRAPHICS ON A CRT:

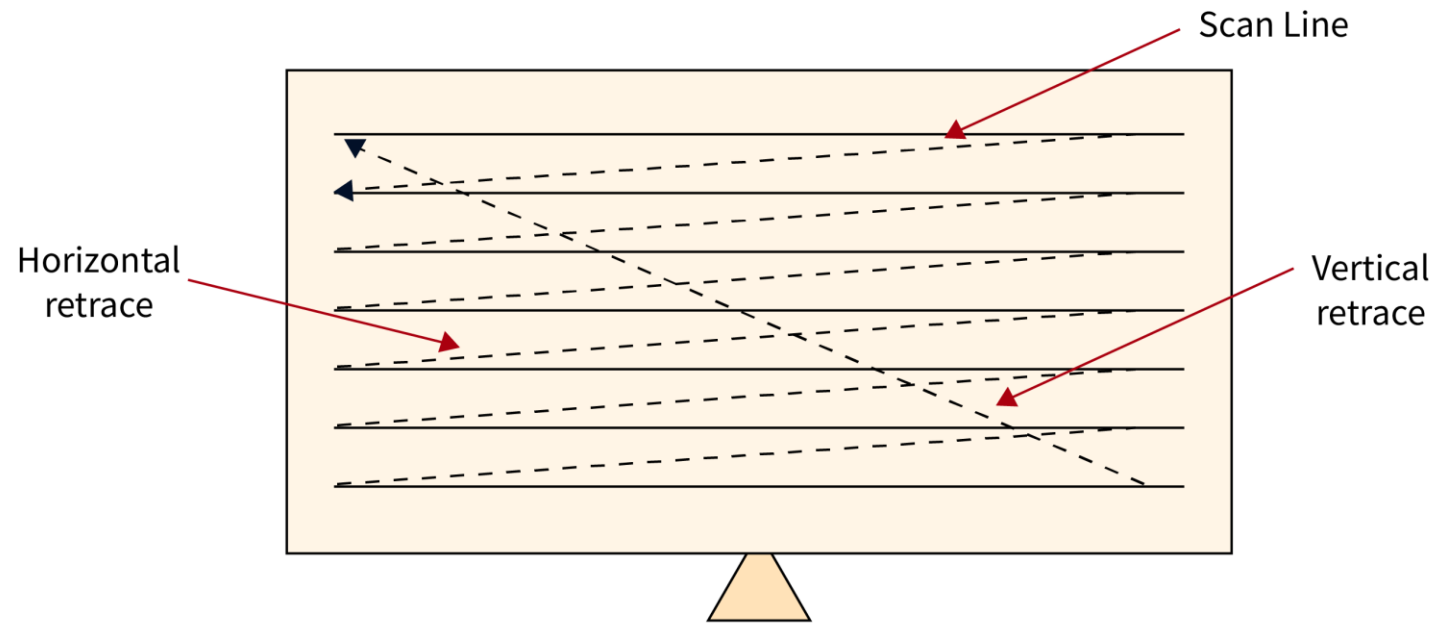
Two classes of computer graphic display *Raster Scan Display* and *Random Scan Display*.



RASTER SCAN DISPLAY

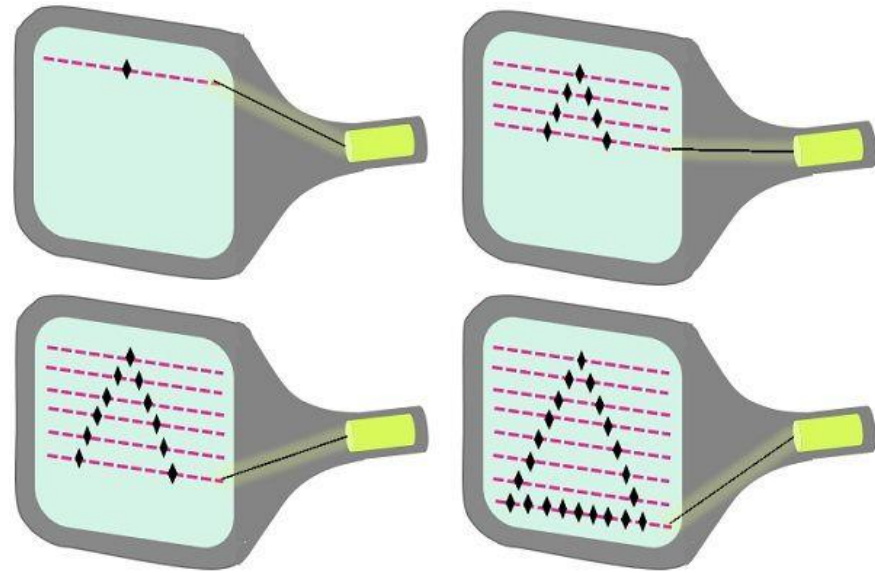
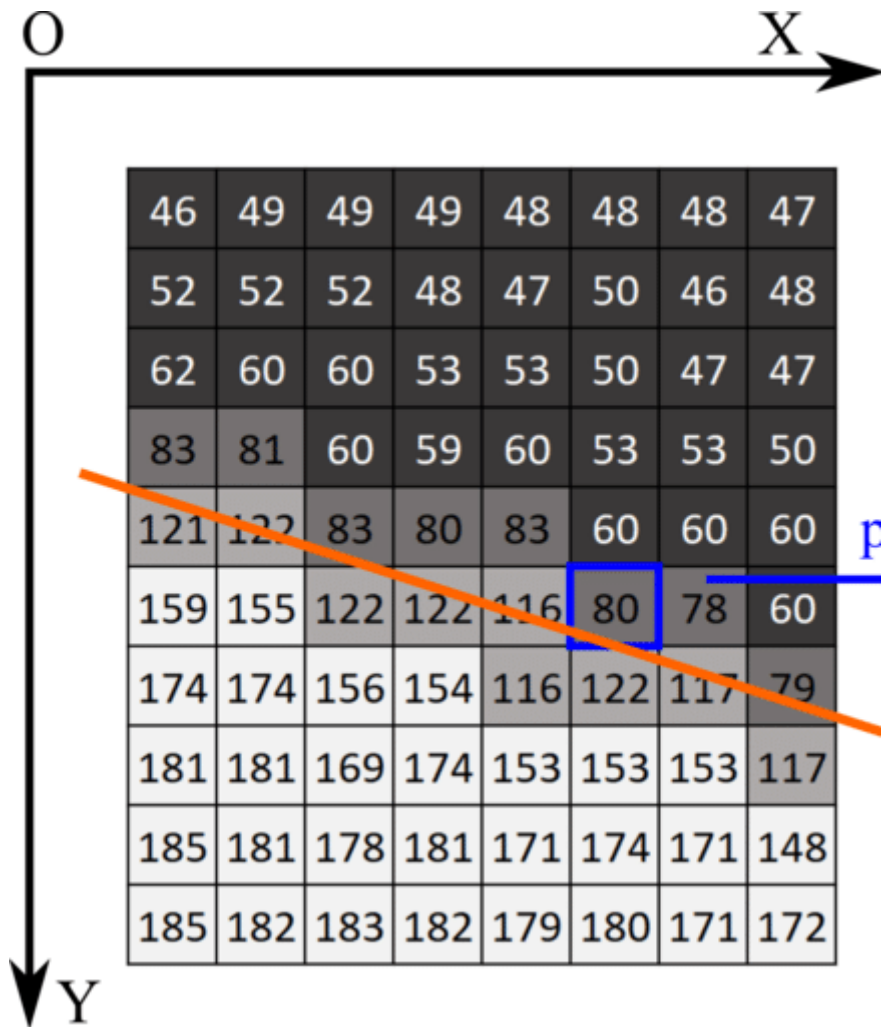
- In this system, the electron beam is swept across the screen **one row at a time from top to bottom.**
- As the electron beam moves across each row, the beam intensity is turned ON and OFF to create a pattern of illuminated spots.
- The picture definition is stored in a memory area called the refresh buffer or frame buffer, which holds the set of intensity values for all the screen points(pixel)
- These values are then retrieved from the refresh buffer and "painted" on the screen one row (scan line) at a time. At the end of each scan line, the electron beam returns to the left side of the screen to begin displaying the next scan line.
- The return to the left of the screen, after refreshing each scan line, is called the **horizontal retrace** of the electron beam. At the end of each frame, the electron beam returns to the top left corner of the screen to begin the next frame. This is known as **vertical retrace**





SCALER
Topics





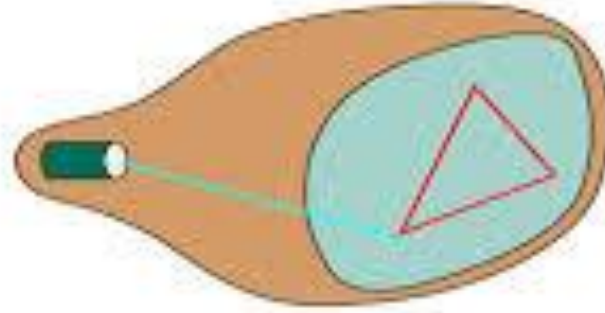
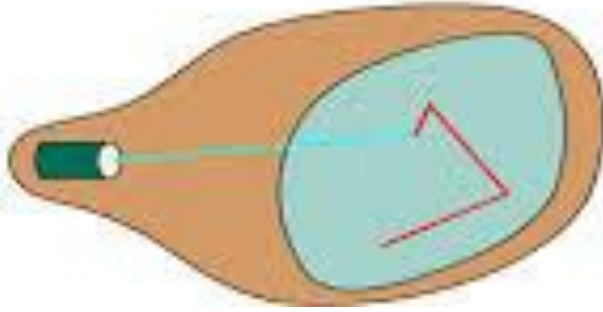
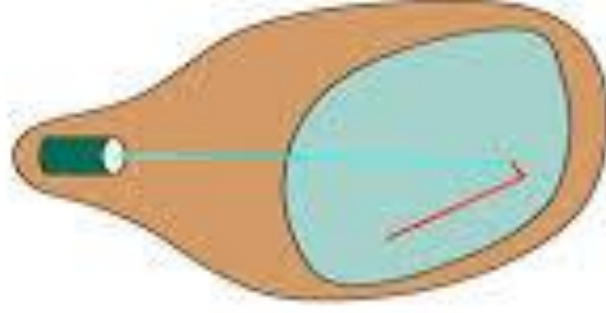
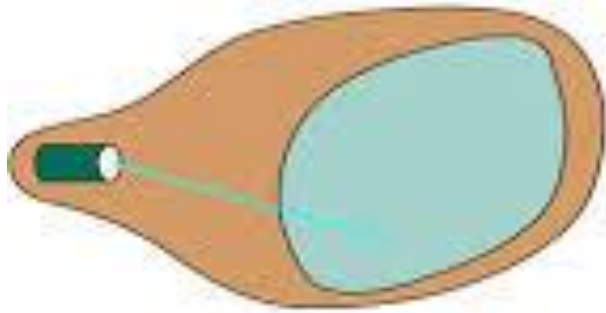
Intensity values: brightness of each pixel represented by a numerical value



RANDOM SCAN DISPLAY

- In this system, a CRT has the electron beam directed only to the parts of the screen where a picture is to be drawn. Random scan monitors draw a picture one line at a time, and for this reason are referred to as vector, stroke writing.
- Picture definition is stored as a set of line-drawing commands in an area of memory referred to as the **refresh display file**.
- Refresh rate on a random scan system depends on the number of lines to be displayed.
- Random scan systems are designed for line drawing applications. They cannot display realistic shaded scenes.





COLOUR DISPLAY ON CRT

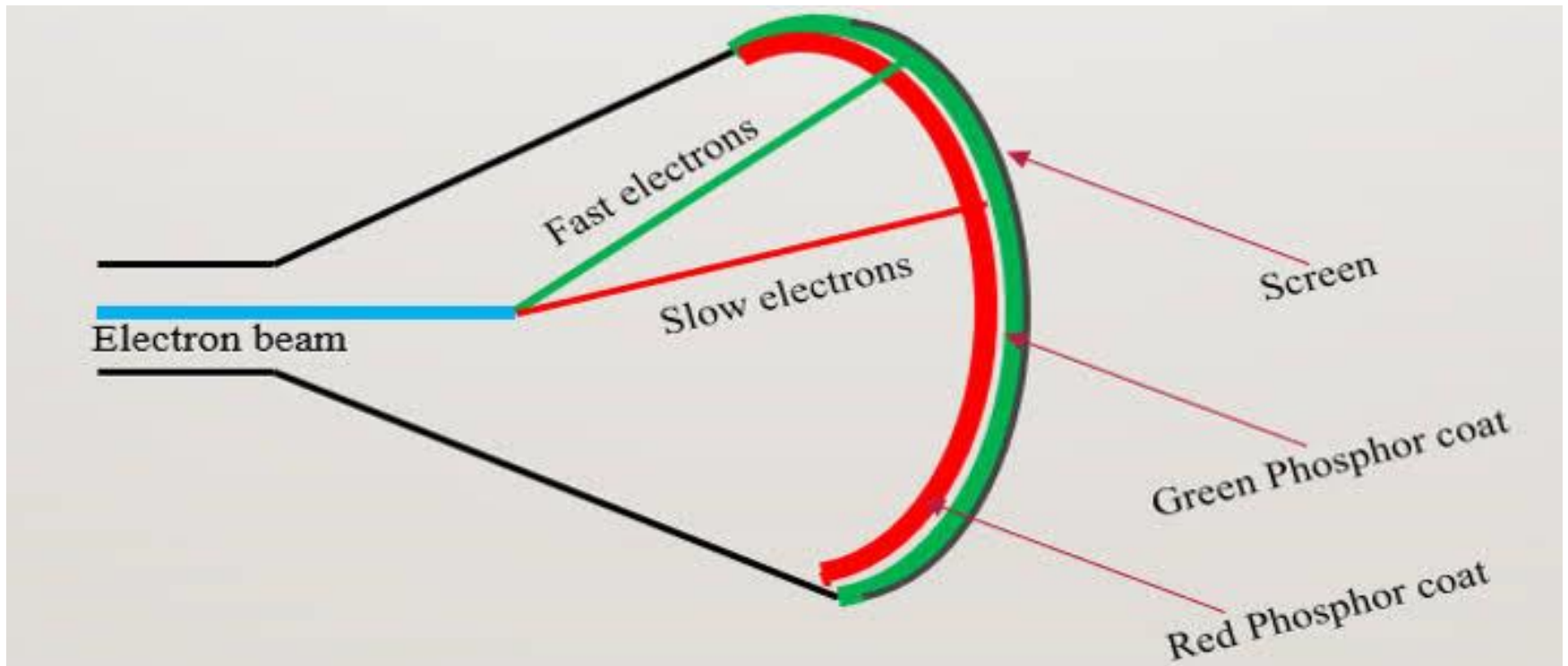
- The two basic techniques for producing colour displays with a CRT are:

Beam Penetration and Shadow Masking

- **Beam Penetration:** In this system, two layers of phosphor (usually red and green) are coated on the inner side of the CRT screen. This produces four colors only, red, green, orange, and yellow. Beam Penetration uses Random Scan System.
 - If a low potential electron beam strikes the face, it excites only the red phosphor and produces red traces.
 - If a high potential electron beam strikes the tube face, it excites the green phosphor and produces green traces.
 - If an electron beam of intermediate potential and velocity, then it produces a combination of red and green i.e., orange and yellow.



Beam Penetration

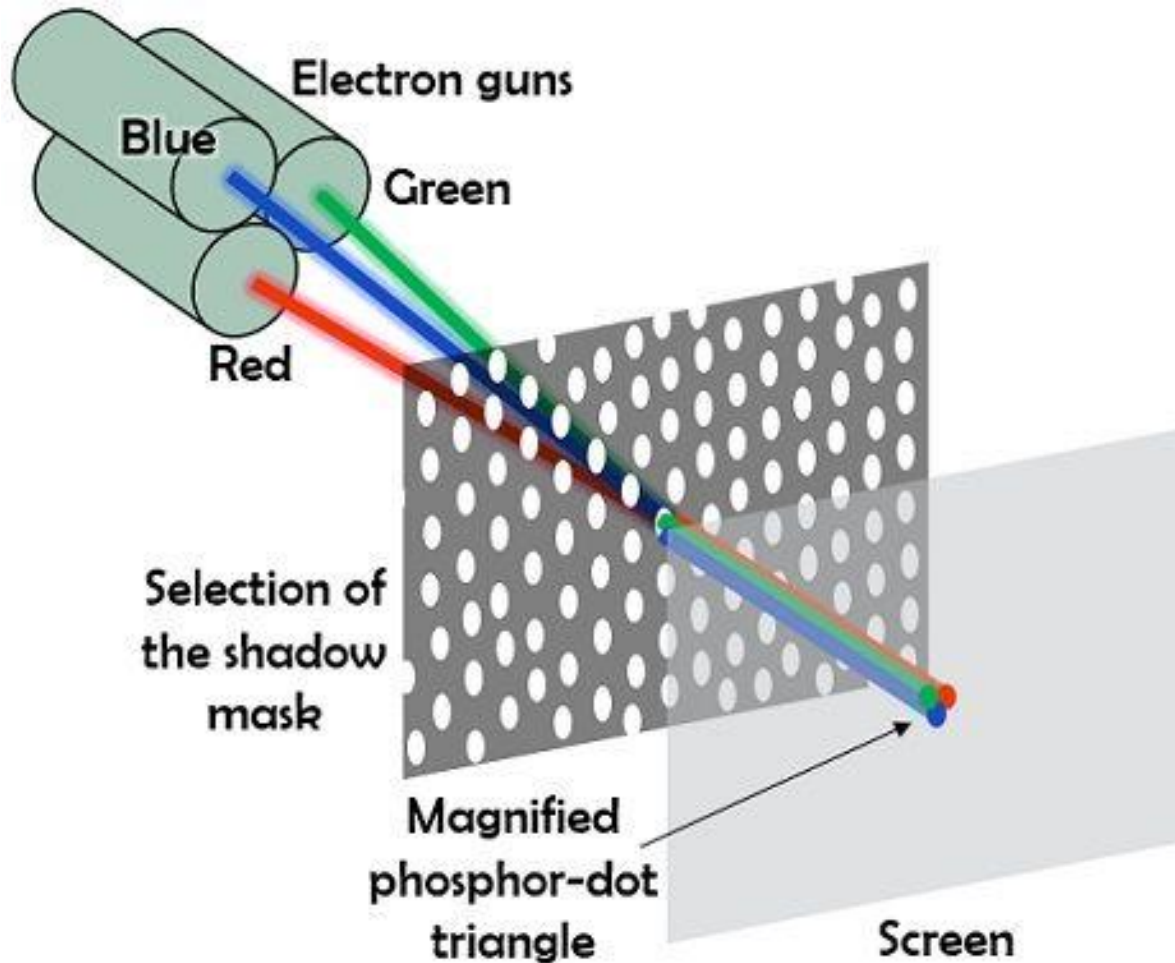


COLOUR DISPLAY ON CRT

- **Shadow Masking:** A shadow-mask CRT has three phosphor colour dots at each pixel position. One phosphor dot emits a red light, another emits a green light and the third emits a blue light.
- This type of CRT has three electron guns, one for each colour dot, and a shadow-mask grid just behind the phosphor-coated screen.
- When the three beams pass through a hole in the shadow mask, they activate a dot triangle, which appears as a small colour spot on the screen.
- Colour variations in a shadow-mask CRT can be obtained by varying the intensity levels of the three electron beams. For example, a white (or grey) area is the result of activating all three dots with equal intensity. Yellow is produced with the green and red dots only,



Shadow Masking



Shadow Masking in Colour CRT



LIQUID CRYSTAL DISPLAY MONITORS (LCD)

- LCD is a combination of two states of matter, the solid and the liquid. The solid part is crystal and this liquid and crystal together make the visible image
- In LCD display, it consists of millions of pixels made of crystal and arranged in a rectangular grid.
- An LCD screen is a collection of multiple layers. **Backlighting, polarizing filters, electrodes, liquid crystal layer, colour filters.**
- A fluorescent light source, known as the backlight, makes up the rearmost layer. Light passes through the first of two polarizing filters.
- The polarized light then passes through a layer that contains thousands of liquid crystal blobs aligned in tiny containers called cells. These cells are aligned in rows across the screen; one or more cells make up one pixel.



LIQUID CRYSTAL DISPLAY MONITORS (LCD)

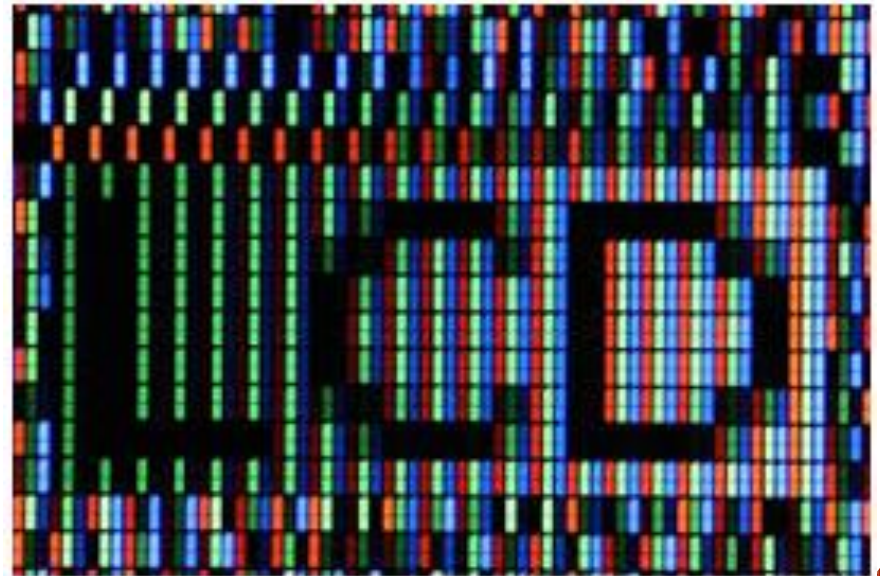
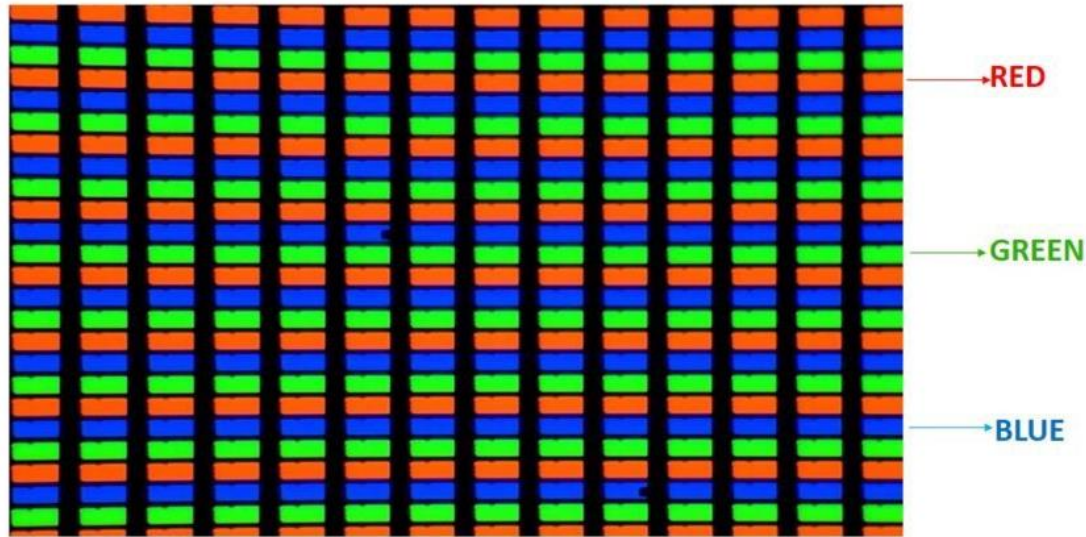
- Electric leads around the edge of the LCD create an electric field that twists the crystal molecule, which lines the light up with the second polarizing filter and allows it to pass through.
- In a coloured LCD panel, each pixel is made up of three liquid crystal cells. In front of each of these cells, there is a red, green or blue filter.
- Light passing through the filtered cells creates the colours on the LCD. Nowadays, nearly every colour LCD uses a thin-film transistor (TFT), also known as an active matrix, to activate each cell.
- LCD address each pixel individually. As a result, it can create sharper text than CRT's



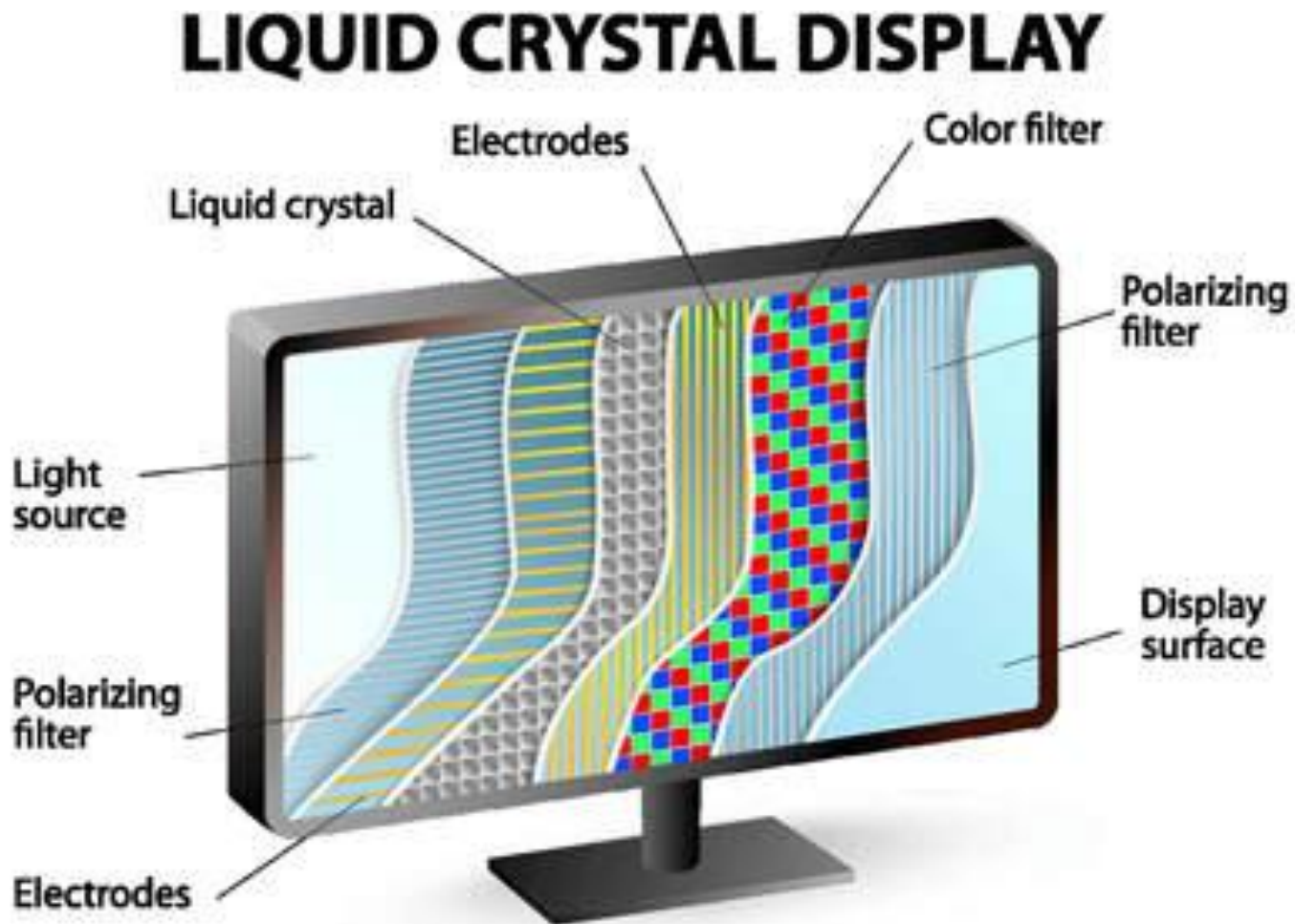
LIQUID CRYSTAL DISPLAY MONITORS (LCD)

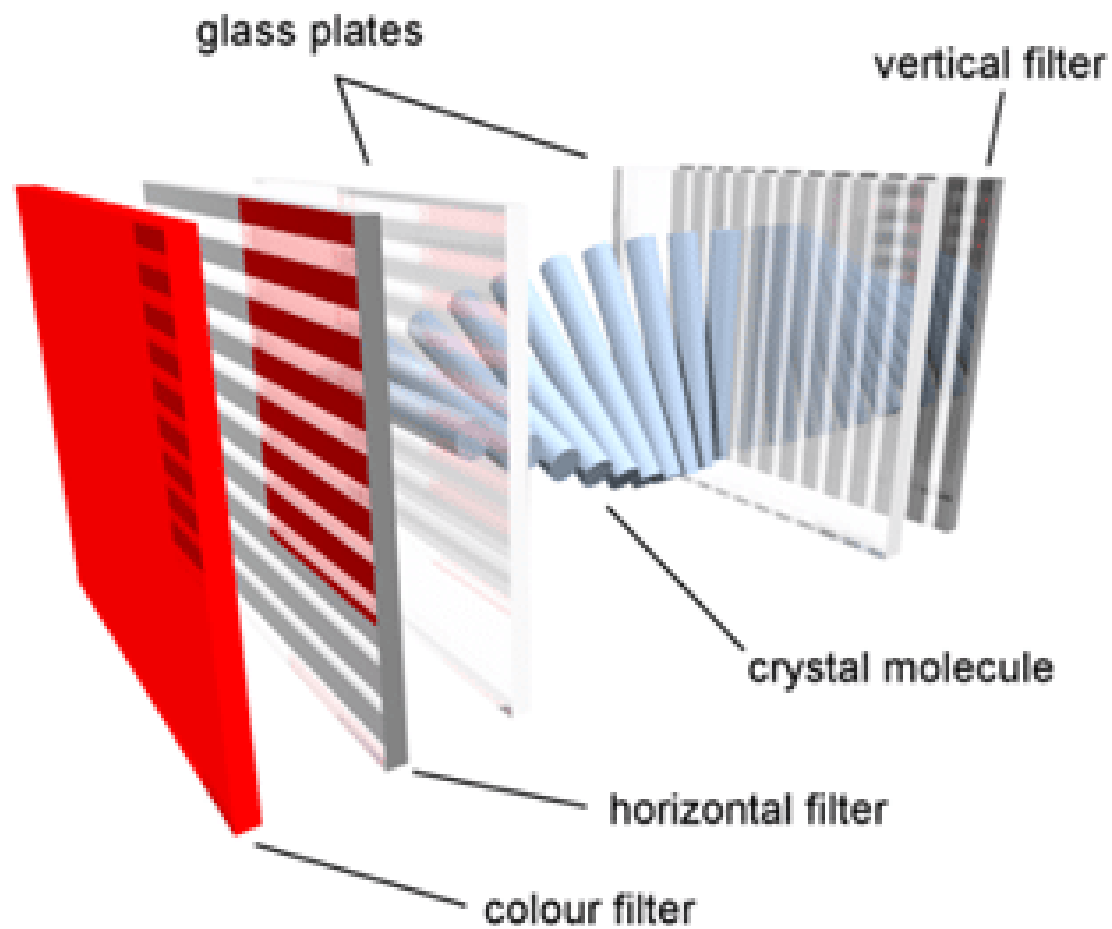


LIQUID CRYSTAL DISPLAY MONITORS (LCD)



LIQUID CRYSTAL DISPLAY MONITORS (LCD)





DIFFERENCES BETWEEN LCD AND CRT:

- **Size:** LCD is lightweight and compact, which saves desktop space as compared to a CRT Resolution
- **Resolution:** LCD is designed to work in a single resolution while CRT is designed for many resolutions.
- **Pixel Density:** Pixel density of LCD is generally not as tight as the dot pitch in CRT but for most applications, the density is acceptable.
- **Brightness:** The illuminated phosphor of a CRT is not nearly as bright as what the LCD can produce with its florescent backlight.
- **Power Consumption:** LCD consumes significantly less power than CRT and has a low emission risk. Typically, a LCD consumes approximately half of the power of a typical CRT.



DIFFERENCES BETWEEN LCD AND CRT:

- **Cost:** Prices for LCD screens are quite high. They are still much costlier than CRT.
- **Viewing Angle:** CRT can be viewed at almost any angle but LCD is best viewed "head on". Even when viewing an LCD head on, narrow viewing angles can appear to have inconsistent colour and brightness.

